

Infosys Springboard Virtual Internship 7.0

Data Visualization

**Cyber Threat Analytics Platform with Incident
Visualization System
(Group 2 – Batch-1)**

Submitted by

Kongi Deepika

1. Project Overview

The Cyber Threat Analytics Platform with Incident Visualization System is a Power BI project developed to analyse incident, threat, location, environmental and response data.

The project helps users understand incident patterns, identify high-risk locations, analyse response performance, assess security risks, identify emerging risks and support management decision-making.

2. Problem Statement

Organizations generate large amounts of incident, threat, operational and environmental data. When this data is spread across different systems, it becomes difficult to identify threats, analyse incident patterns and make quick decisions.

The Cyber Threat Analytics Platform with Incident Visualization System brings this information together into an interactive Power BI solution. It helps analyse incidents, identify risk areas, evaluate response performance, monitor emerging risks and support better decision-making.

3. Project Objectives

The project aims to:

- Analyse threat categories and severity.
- Identify incident hotspots.
- Analyse environmental and temporal factors.
- Evaluate incident response performance.
- Assess security risks.
- Identify resource pressure and security weaknesses.
- Monitor emerging risks.
- Forecast incident trends.
- Provide executive-level insights.

4. Data Preparation and Modelling

The project began with data preparation and transformation using Power Query in Power BI.

Activities performed

- Removed unnecessary fields.
- Checked data types.
- Cleaned invalid and inconsistent values.
- Prepared incident information.
- Organised location information.

- Prepared environmental/weather information.
- Prepared response and operational information.
- Created relationships between relevant tables.
- Created calculated columns where required.
- Developed reusable DAX measures.

The prepared data was then used to create interactive dashboards.

5. Dashboard Implementation

5.1 Threat Classification & Severity Analytics Dashboard

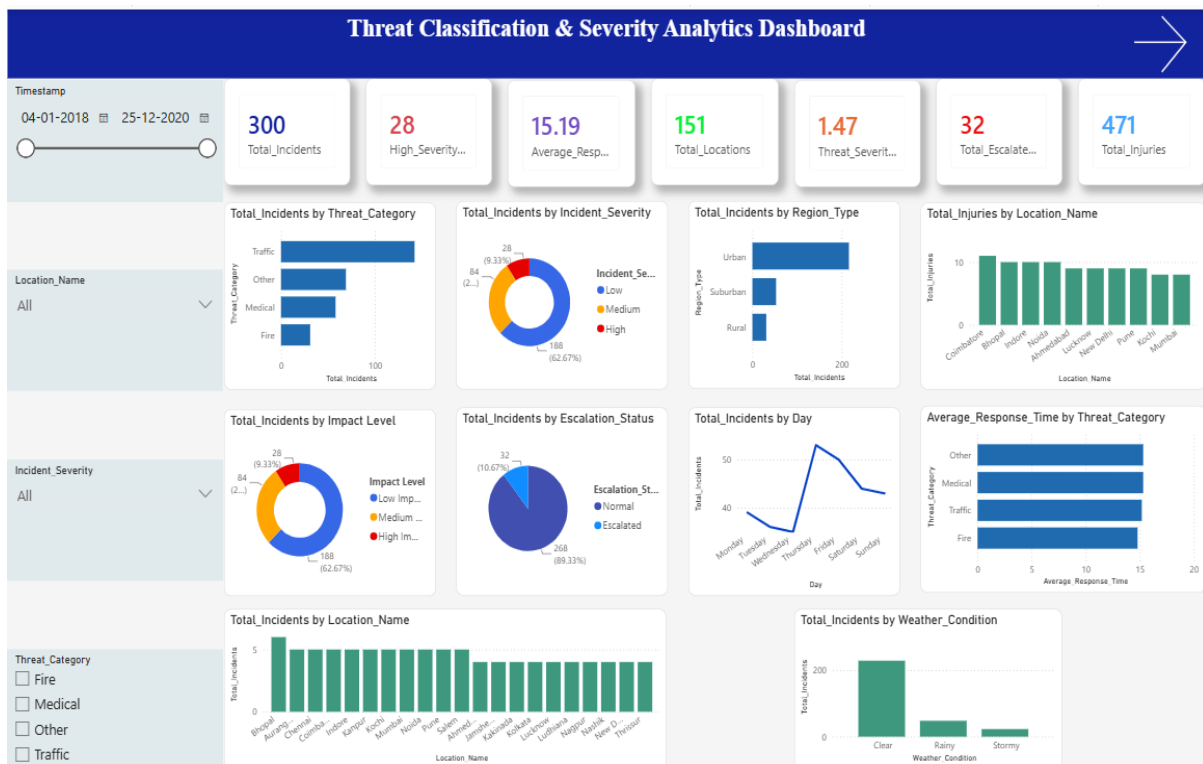
This dashboard provides the basic understanding of the threat environment by analysing incident categories and severity.

What it provides

The dashboard helps identify:

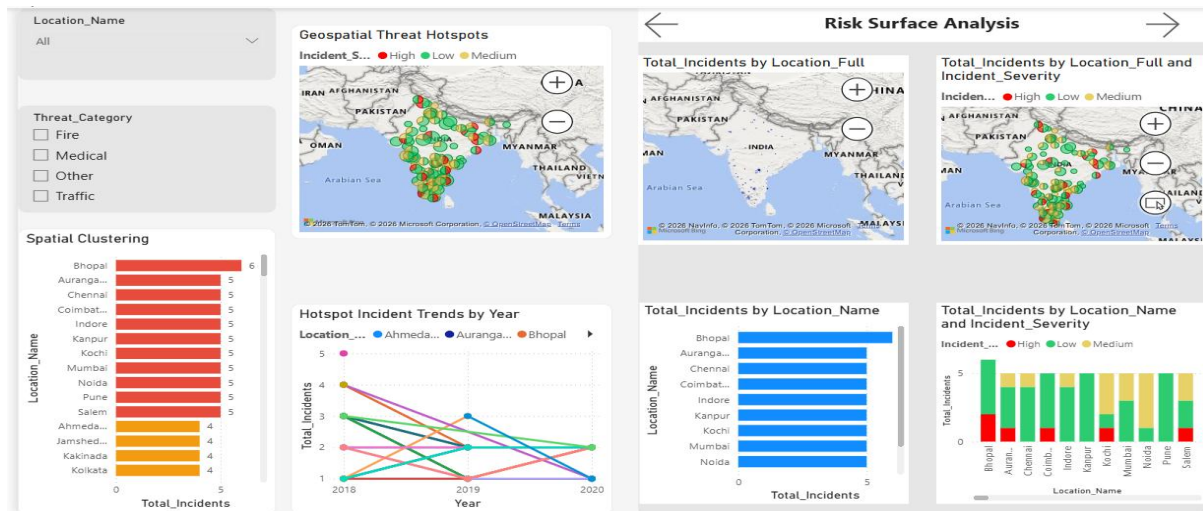
- Most common threat categories.
- Severity distribution.
- Locations with high incident activity.
- Changes in incident activity over time.

Dashboard :



5.2 Geospatial Intelligence Dashboard

The Geospatial Intelligence dashboard identifies where incidents occur and highlights geographical hotspots.



Location Tooltip

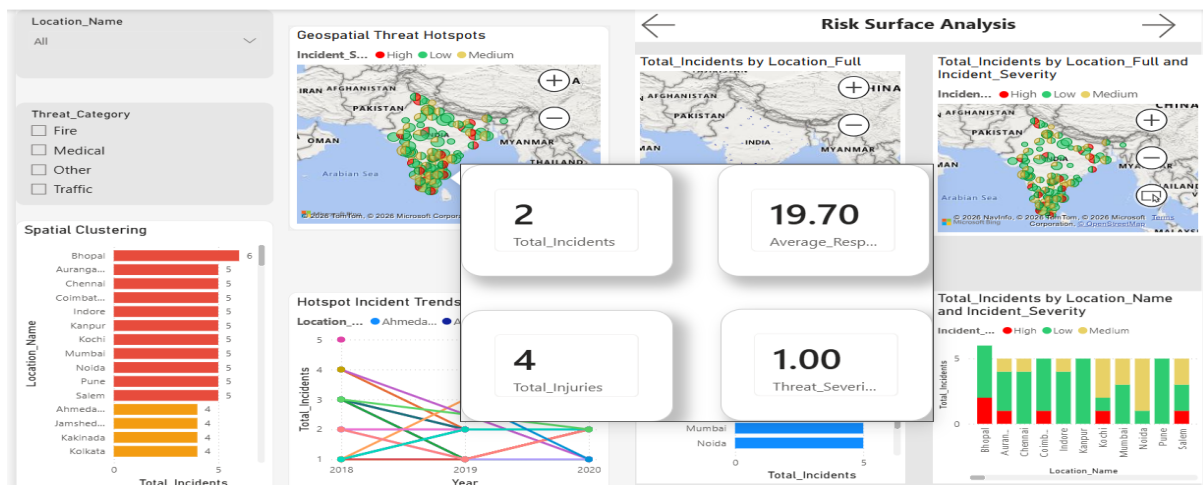
A custom tooltip provides additional information when the user hovers over a location.

The tooltip includes:

- Total Incidents
- Threat Severity Index
- Average Response Time
- Total Injuries

What it provides

It helps users quickly understand the incident severity, response performance and overall impact of a location without leaving the Geospatial Intelligence page.



5.3 Environmental Factors & Effective Analytics

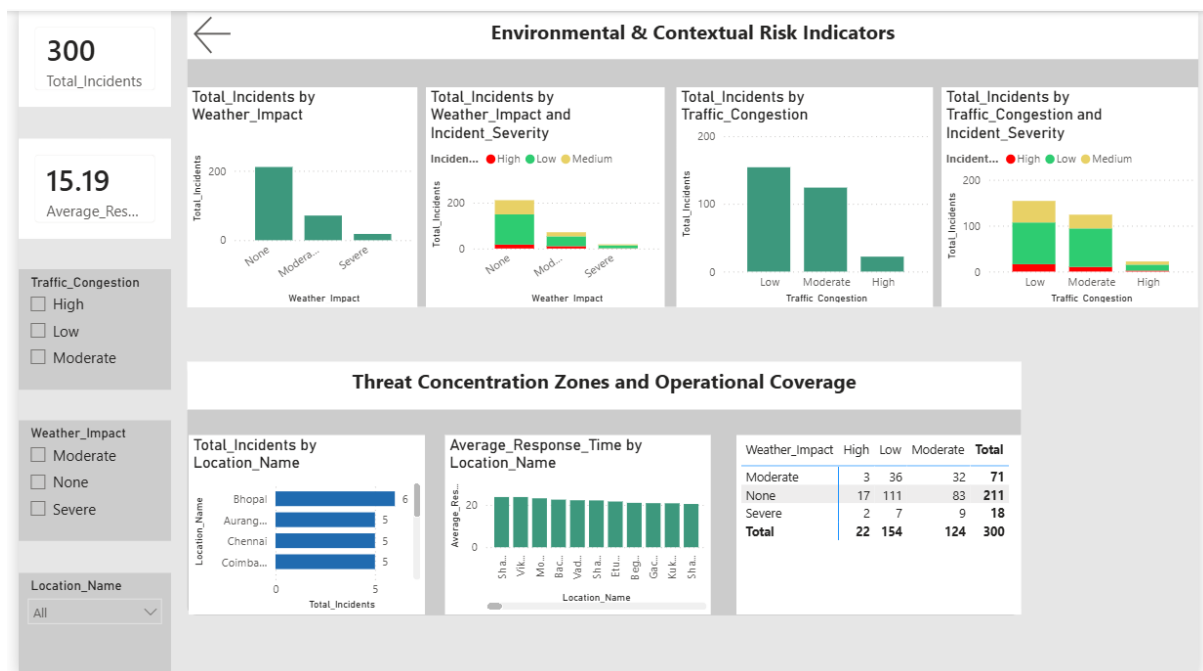
This dashboard is a continuation of the Geospatial Intelligence analysis.

It analyses environmental and situational factors associated with incident activity and operational performance.

What it provides

The dashboard helps stakeholders understand how incident activity and response performance vary under different environmental conditions.

Dashboard :



5.4 Location Details Dashboard

The Location Details page provides detailed information about a selected location.

It is mainly accessed through the Drill-through feature.

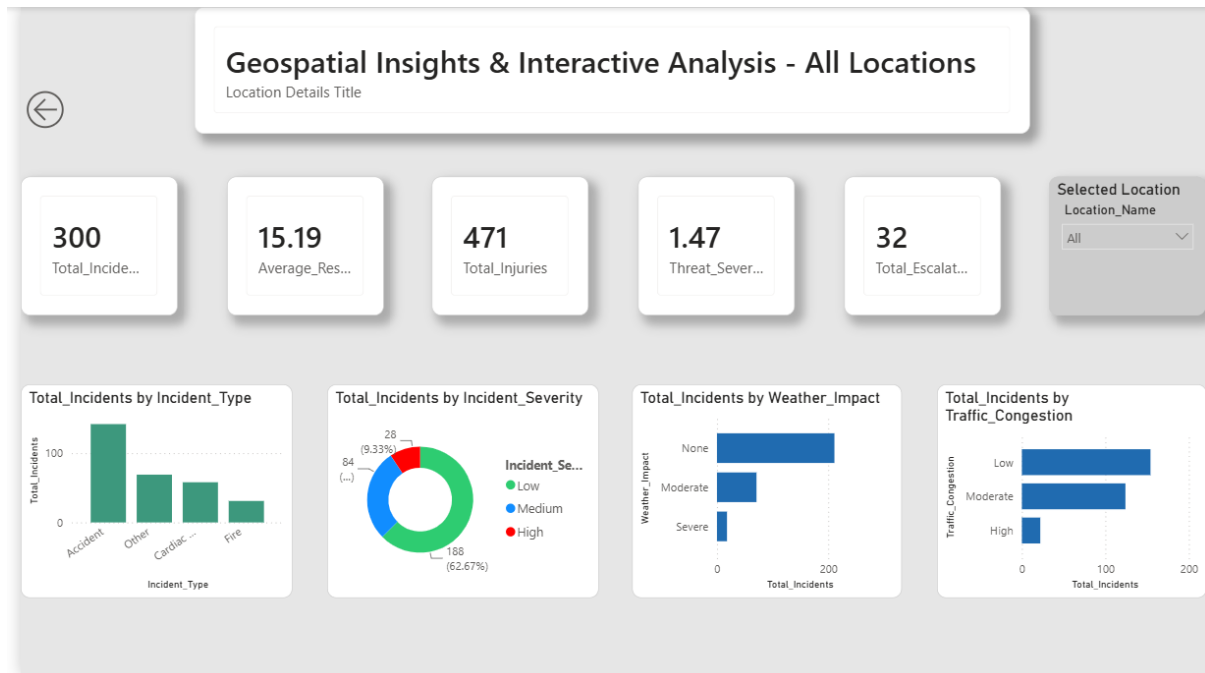
Drill-through

Users can select a location from the Geospatial dashboard and navigate to the Location Details page.

Geospatial Dashboard → Select Location → Drill-through → Location Details

This provides deeper analysis without overcrowding the main dashboard.

Dashboard :



5.5 Location Tooltip

The Location Tooltip provides additional information about geographical points without adding unnecessary information to the main map.

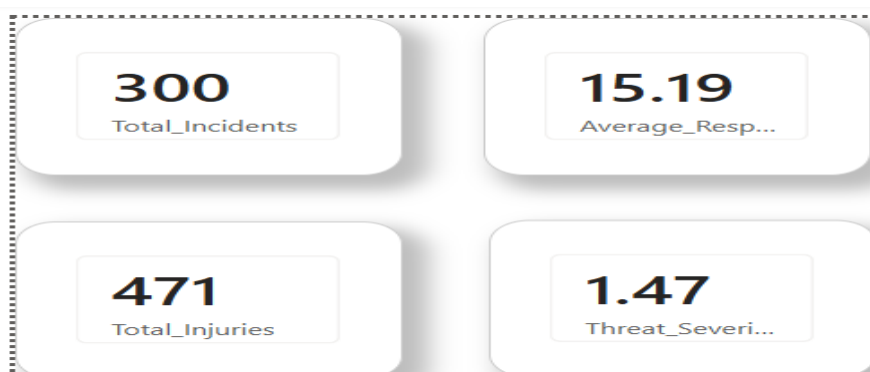
Information displayed

- Total Incidents
- Threat Severity Index
- Average Response Time
- Total Injuries

Benefit

Users can obtain detailed information simply by hovering over a location on the map.

Tooltip :



5.6 Temporal Incident Analytics Dashboard

The Temporal Incident Analytics dashboard analyses when incidents occur and identifies changing incident patterns.

Key analysis

Trend by Severity shows how different severity levels change over time.

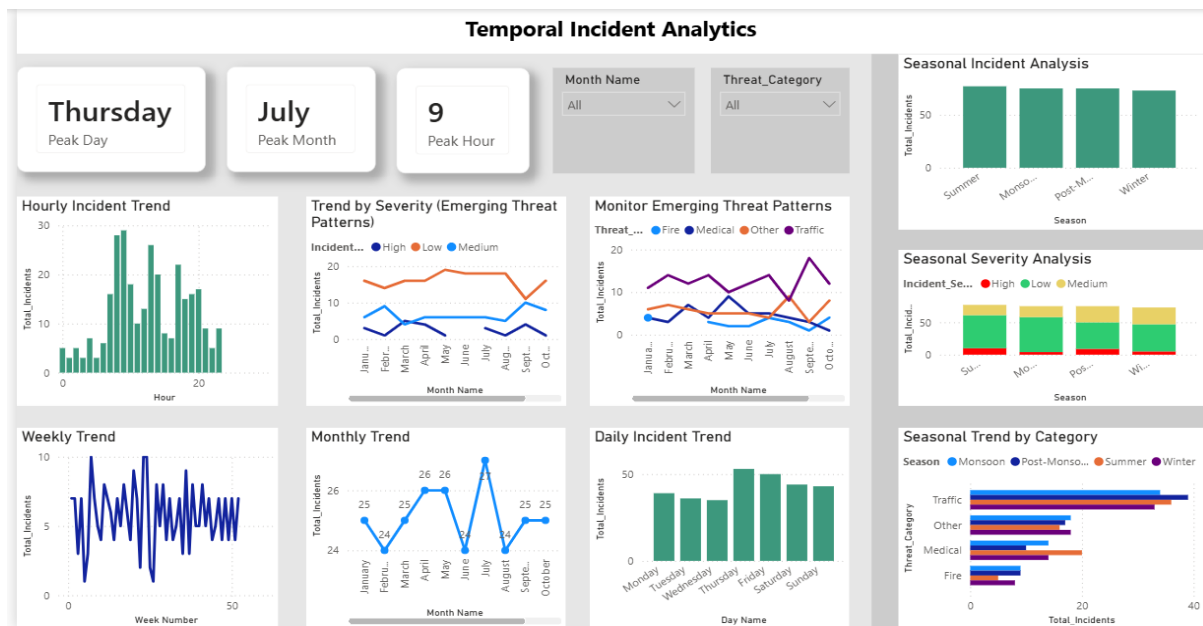
Emerging Threat Pattern helps identify threat categories whose activity is increasing or changing over time.

What it provides

The dashboard helps identify:

- Peak incident periods.
- Temporal patterns.
- Severity changes.
- Emerging threat patterns.

Dashboard :



5.7 Correlation Analysis Dashboard

The Correlation Analysis page is a continuation of the Temporal Incident Analytics module.

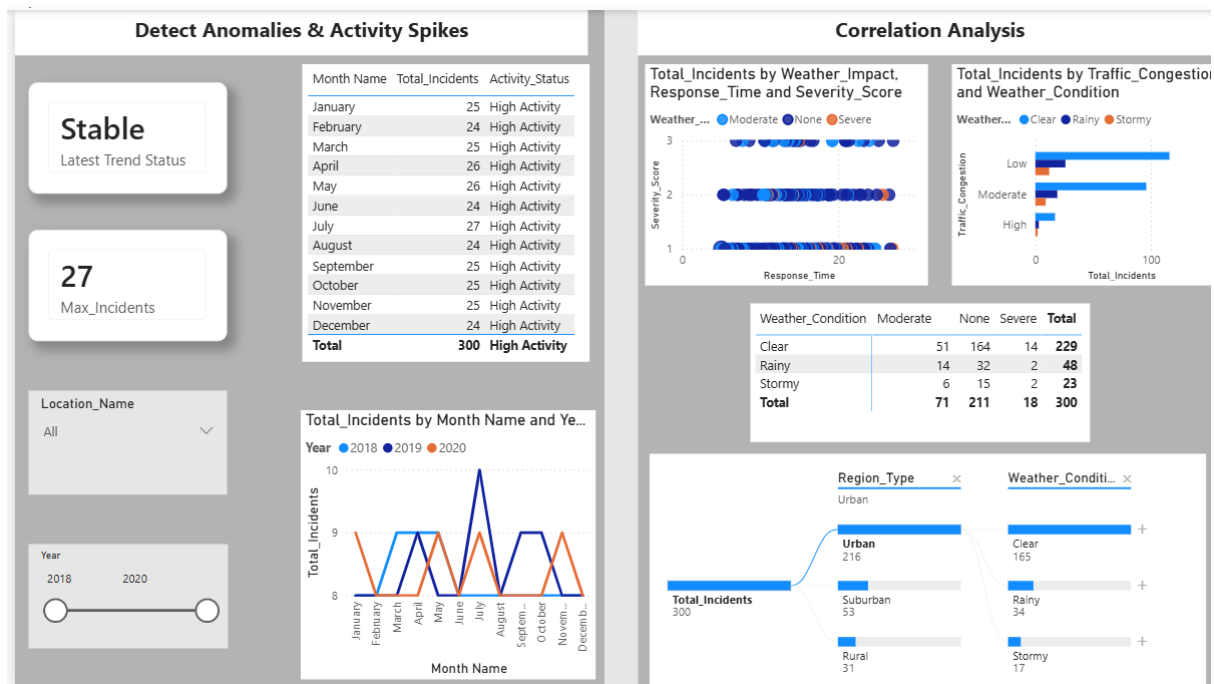
It examines relationships between incident, environmental and operational variables.

What it provides

The dashboard helps identify relationships between different factors.

Correlation is used to identify association, not to claim causation.

Dashboard :



5.8 Incident Response Analytics Dashboard

This dashboard evaluates the effectiveness of incident response operations.

Operational Unit

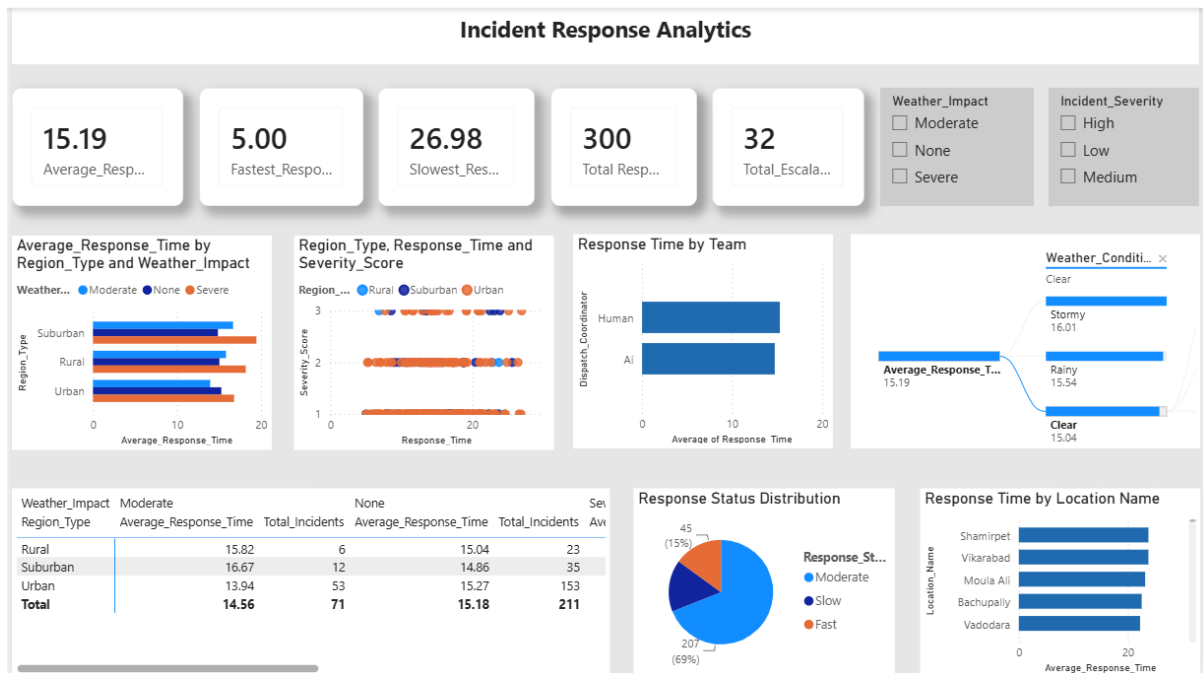
Dispatch Coordinator is used as the operational-unit field because it represents the available operational coordination information in the existing response data.

What it provides

The dashboard helps identify:

- Response bottlenecks.
- Locations with slower responses.
- Differences in response performance.
- Operational areas requiring improvement.

Dashboard :



5.9 Security Risk Assessment Dashboard

This dashboard evaluates the organization's current security exposure and operational vulnerabilities.

It addresses:

- Threat Exposure Level Analysis.
- Operational Vulnerability Analysis.
- Risk Indicators by Location.
- Risk Indicators by Operational Unit.
- Resource Allocation Pressure.
- Security Weakness Analysis.
- Security Risk Scorecard.

Risk Classification

Risk Score	Risk Level
≥ 40	High Risk
$20 < 40$	Medium Risk
< 20	Low Risk

What it provides

Management can identify high-risk locations and operational areas requiring additional monitoring or resource attention.

Dashboard :



5.10 Predictive Intelligence Dashboard

The Predictive Intelligence dashboard extends the analysis from historical information towards current and future risk assessment.

Requirements addressed

- Predictive Threat Risk Model.
- Incident Risk/Likelihood Analysis.
- Predicted Risk Table
- Forecasting of Incident Trends.
- Potential High-Risk Zones.
- Current & Emerging Risk Zones

Current & Emerging Risk Zones

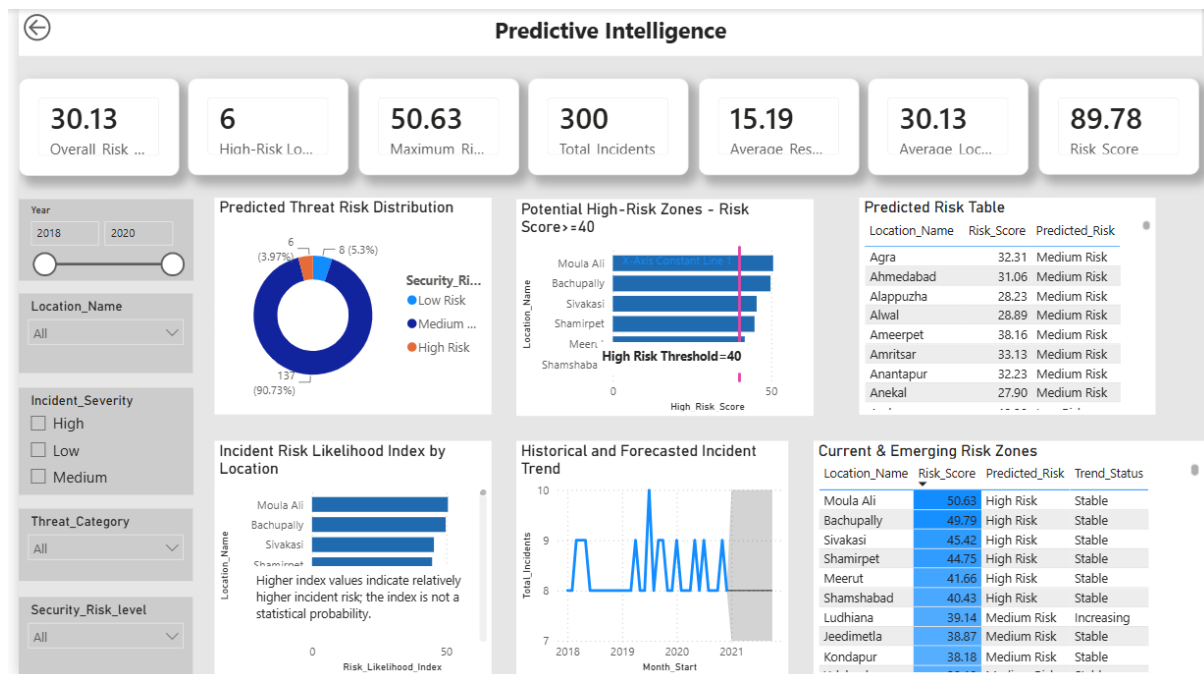
The table contains:

- Location
- Risk Score
- Predicted Risk
- Trend Status

An Increasing trend represents increasing historical incident activity. It is not itself a statistical probability.

The forecast line chart is therefore used separately to represent forecasted incident trends.

Dashboard :



5.11 Executive Intelligence Dashboard

The Executive Intelligence dashboard provides management with a concise overview of current security conditions and emerging risks.

KPIs

- Overall Risk Score
- High-Risk Locations
- High-Risk Incidents
- Maximum Risk Score
- Average Response Time
- Total Escalated Incidents

Visuals

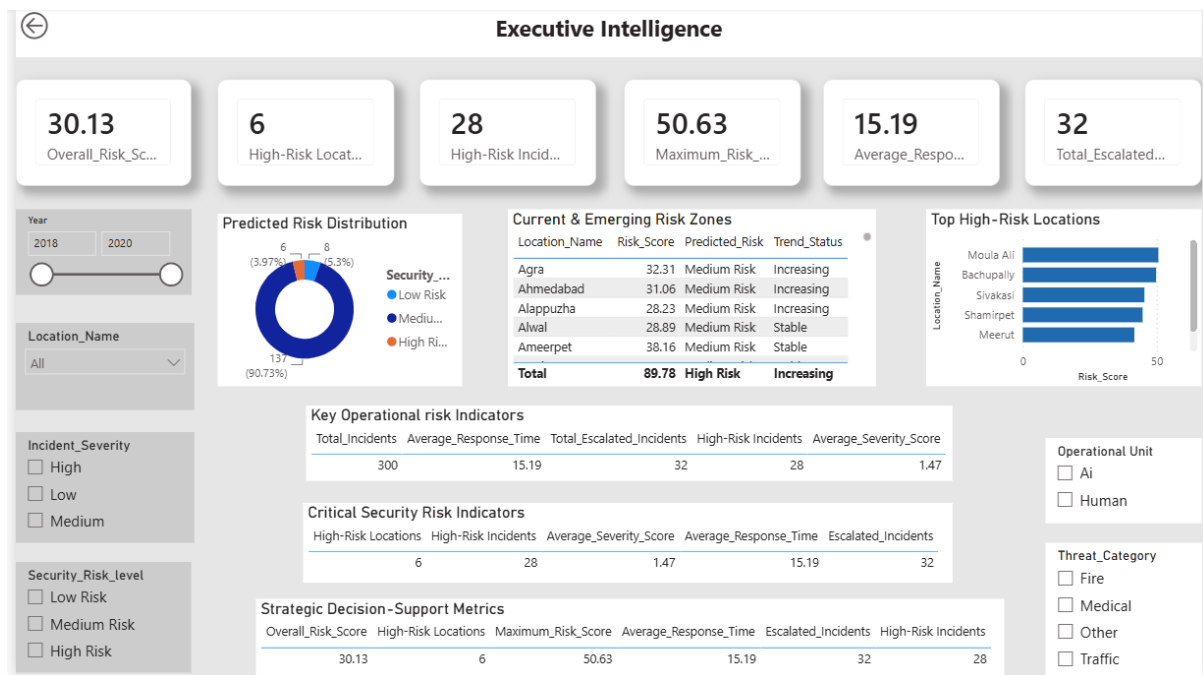
- Predicted Risk Distribution
- Top High-Risk Locations
- Current & Emerging Risk Zones

- Critical Security Risk Indicators
- Strategic Decision -support Metrics
- Key Operational Risk Indicators

What it provides

The dashboard converts detailed operational and risk analytics into concise information for strategic decision-making.

Dashboard :



6. Power BI Features Implemented

Sync Slicers

Sync Slicers were used to maintain consistent filtering across relevant dashboard pages.

Important slicers include:

- Year
- Location/City
- Severity
- Category
- Operational Unit
- Risk Level

This allows users to apply a filter consistently across multiple analytical pages.

Edit Interactions

Edit Interactions were used to control how visuals interact with each other.

For example:

- Selecting a category updates related visuals.
- Selecting a location updates location-related charts.
- Selecting severity updates risk and response visuals.
- Selecting a map point filters relevant analytical visuals.

This prevents unwanted interactions and improves dashboard usability.

Drill-through

The Drill-through feature connects summary and detailed analysis.

Example:

Geospatial Intelligence → Location → Drill-through → Location Details

This allows users to investigate a selected location in greater detail.

Page Navigation

Navigation features were used to connect the different analytical pages.

The dashboard flow is:

Threat → Geospatial → Environment → Location → Temporal → Correlation → Response → Risk → Predictive → Executive

This makes the report easier to navigate and presents the project as one integrated analytics platform.

Tooltips

Tooltips provide additional information when users hover over visual elements.

The Location Tooltip is particularly useful for the geographical map because it provides detailed risk and incident information without overcrowding the map.

7. Technologies Used

Technology	Purpose
Microsoft Power BI	Dashboard and visual analytics
Power Query	Data cleaning and transformation
DAX	Measures, KPIs and risk calculations
Power BI Data Model	Data relationships and analytical structure
Power BI Maps	Geographical analysis
Power BI Forecasting	Incident trend forecasting
Decomposition Tree	Detailed factor analysis
Sync Slicers	Consistent filtering
Drill-through	Detailed location investigation
Edit Interactions	Visual interaction control
Tooltips	Additional contextual information
Page Navigation	Navigation between dashboards

8. Project Uniqueness

The uniqueness of the Cyber Threat Analytics Platform with Incident Visualization System lies in its integrated approach.

The platform combines:

- Threat classification.
- Severity analysis.
- Geographical intelligence.
- Environmental analysis.
- Temporal analytics.
- Correlation analysis.
- Incident response analytics.
- Security risk assessment.
- Resource allocation pressure.
- Predictive intelligence.
- Executive decision support.

It also provides advanced Power BI interaction features such as:

- Sync Slicers.
- Drill-through.
- Edit Interactions.
- Page Navigation.
- Tooltips.
- Cross-filtering.

This makes the platform more than a static reporting dashboard; it acts as an interactive security analytics environment.

Key Insights

1. High-risk incidents are concentrated in a small number of locations.
2. Incident severity and response performance vary across locations.
3. Some locations show increasing historical incident trends.
4. Resource pressure is higher where incident volume is high relative to available resources.
5. Predictive risk analysis highlights locations requiring closer monitoring.

Strategic Recommendations

1. **Prioritise high-risk locations:** Allocate additional monitoring and response resources to locations with consistently high risk scores.
2. **Improve resource allocation:** Review resource availability in operational areas with higher incident pressure and slower response times.
3. **Monitor emerging risks:** Closely monitor locations with increasing historical trends and regularly review forecasted incident trends.

9. Conclusion

The Cyber Threat Analytics Platform with Incident Visualization System successfully integrates incident analytics, geographical intelligence, environmental analysis, temporal analysis, response performance, security risk assessment and predictive intelligence into a unified Power BI environment.

The solution provides different levels of analysis for different users. Operational users can investigate incidents and locations in detail, while management can use the Security Risk

Assessment and Executive Intelligence dashboards to understand current risks and make informed decisions.

The implementation of Sync Slicers, Drill-through, Edit Interactions, Page Navigation, Tooltips further improves the usability and effectiveness of the dashboards.

Overall, the project provides an end-to-end approach for moving from historical incident analysis to current risk assessment and future-oriented security intelligence.